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RESEARCH ARTICLE

Stochastic Valuation of Barrick Mining Corporation

An integrated option-implied and Monte Carlo valuation architecture

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Abstract

This study develops and audits an integrated stochastic valuation framework for Barrick Mining Corporation within Starting Finance Club PoliTo Research. The architecture combines probabilistic modelling, econometric analysis, Monte Carlo simulation, historical gold-price dynamics, stochastic volatility and jump processes, option-implied calibration, purpose-built mine operating forecasts, and discounted cash flow valuation. Independent research modules are unified through explicit source ownership and consistent date, unit, measure, and accounting conventions. Alternative stochastic specifications are applied exclusively to the gold-price layer, while production and cost forecasts remain fixed across valuation engines through common operating-model inputs. Taxation, reinvestment, cost of capital, terminal value, and the equity bridge are likewise shared across scenarios. Option-market data are used to calibrate gold-price dynamics under the risk-neutral measure, allowing the effect of alternative distributional assumptions to be examined within an otherwise common corporate valuation contract. Results indicate that model choice can alter both central valuation outcomes and tail behaviour, while the observed market price remains above the medians generated by all four specifications. The resulting distributions are therefore interpreted as controlled stochastic valuation sensitivities rather than physical forecasts, confidence intervals, fair-value estimates, price targets, or investment recommendations.

RESEARCH PROGRAMME AND COLLABORATION

This paper grew out of an eight-team project involving 17 students with complementary backgrounds in mathematics, data science, engineering, economics and finance. The size of the author group reflects a division of research responsibilities and a shared learning process: Teams 1–3 developed the mathematical, econometric and simulation foundations; Teams 4–5 built the operating and corporate-valuation layers; and Teams 6–8 studied gold-price dynamics, market evidence and option-implied calibration.

The long-form *UNIFICATO* version records definitions, derivations and implementation choices to establish a common technical language across these backgrounds. Teams worked on dedicated modules, then compared assumptions, revised their contributions and reconciled dates, units and interfaces before integration. The present paper condenses this work into a single auditable experiment. Together with the reproducible code, it demonstrates the group's existing research practice and provides a basis for developing a sustained student-led quantitative research team with academic guidance.

1 INTRODUCTION

A gold producer's value depends on the distribution of future commodity prices as well as its operating and financing assumptions. A single price path can hide this source of model risk. We ask how Barrick Mining Corporation's conditional equity-value distribution changes when progressively richer gold-price processes are substituted within a common operating forecast and discounted-cash-flow contract.

We compare Black–Scholes, Heston, Bates–Poisson and Full Bates–Hawkes calibrated to GLD options. Current-surface diagnostics and rolling out-of-sample tests distinguish fit from predictive performance; shared production, costs and WACC shocks isolate the effect of the gold engine. The contribution is this controlled comparison and its auditable integration, rather than a claim that any model provides an unconditional fair value.

2 DATA AND VALUATION SETUP

2.1 Controlled layer separation

The integrated experiment is organized as a controlled handoff across three modules. Team 8 governs only the gold-price process; Team 4 supplies the operating forecasts; and Team 5 maps those inputs through a common tax, WACC, terminal-value and equity-bridge contract. In the four-model valuation comparison, only the gold engine changes. Barrick's Q1–Q2 2026 actuals and the same Team 4 production and unit-cost forecasts from Q3 2026 onward are frozen over 20 quarters; the same Team 5 corporate rules and WACC shocks are then reused across runs.

2.2 Calibration snapshot and measure discipline

The current Team 8 calibration uses the 2 September 2026 GLD call surface. After the common DTE $\geq$ 75 and numerical filters, the dense surface contains 605 eligible observations across 12 expiries and 146 distinct strikes, with 79–653 days to expiry. The official 8×8 Chebyshev–Chebyshev sample contains 64 distinct actual contracts spanning eight expiries and 20 strikes. The risk-free input is a same-date NSS fit to 14 U.S. Treasury tenors. Its 2.0608 bp RMSE is computed against the continuously compounded par-yield proxy; the curve is not described as a bootstrapped zero curve. The Barrick market reference is the 2 September close of USD 44.13, while the common gold anchor remains Barrick's Q2 2026 average realized price of USD 4,417/oz. Only the level anchor is therefore asynchronous with the option and rate snapshot.

The option layer is estimated under the risk-neutral measure $\mathbb{Q}$. In the corporate experiment, only the option-implied distributional shape is transferred to the realized-price anchor. No one-for-one GLD-share-to-gold-ounce conversion and no validated $\mathbb{Q} \to \mathbb{P}$ transformation are claimed.

Table 1. Frozen contract used for model attribution.

Block	Current experiment
Gold layer	Black–Scholes/GBM, Heston, Bates–Poisson and Full Bates–Hawkes calibrated by Team 8 under $\mathbb{Q}$.
Operations	Barrick Q1–Q2 2026 actuals followed by Team 4 deterministic production and unit-cost vectors from Q3 2026.
Corporate mapping	Team 5 tax, growth–ROIC–reinvestment schedule, stochastic WACC, terminal-value and equity-bridge logic.
Simulation	8,192 paths, 260 fine steps, 20 quarters and a common WACC shock array.

2.3 Operating-model status

Team 4's original operating study is richer than the object ultimately passed downstream. Cost of Sales is forecast from a log-space master chain; production is constructed bottom-up from ore processed, average grade and recovery. The unified valuation refreshes the handoff with 2026 actuals but still freezes the resulting 20-quarter production and cost vectors. Consequently, mine-level operating uncertainty is not propagated jointly with gold-price uncertainty. Team 4's legacy price simulation and illustrative EBITDA/valuation output are explicitly excluded.

3 METHODOLOGY

3.1 Gold-price engines

The four Team 8 specifications relax one restriction at a time. Under $\mathbb{Q}$, Black–Scholes/GBM uses constant volatility,

$$\frac{\mathrm{d}S_t}{S_t} = (r_t - q_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t. \tag{1}$$

This constant-volatility specification also corresponds to the earlier internal working-paper draft that paired Team 4 operating forecasts with a provisional market-implied gold-price bridge based on GLD volatilities and a Nelson–Siegel–Svensson Treasury curve. In the present paper it is retained as the simplest benchmark inside the broader four-engine comparison. Heston introduces mean-reverting stochastic variance (Heston, 1993),

$$\frac{\mathrm{d}S_t}{S_t} = (r_t - q_t)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S, \tag{2}$$

$$\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v, \qquad \mathrm{d}W_t^S\,\mathrm{d}W_t^v = \rho\,\mathrm{d}t. \tag{3}$$

Bates adds compensated compound-Poisson jumps (Bates, 1996),

$$\frac{\mathrm{d}S_t}{S_{t^-}} = (r_t - q_t - \lambda k)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S + (e^J - 1)\,\mathrm{d}N_t, \tag{4}$$

where $k = \mathbb{E}^{\mathbb{Q}}[e^J - 1]$.

3.2 Full Bates–Hawkes: implemented affine engine

The full engine preserves Heston variance but replaces the constant Poisson arrival rate with an exponential Hawkes state (Hawkes, 1971):

$$\frac{\mathrm{d}S_t}{S_{t^-}} = (r_t - q_t - \lambda_t k)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S + (e^J - 1)\,\mathrm{d}N_t, \tag{5}$$

$$\mathrm{d}v_t = \kappa(\theta_v - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v, \tag{6}$$

$$\mathrm{d}\lambda_t = \beta(\bar{\lambda} - \lambda_t)\,\mathrm{d}t + \alpha\,\mathrm{d}N_t, \qquad n = \alpha/\beta < 1. \tag{7}$$

Under the implemented exponential-kernel specification, pricing exploits the affine Markov structure rather than a standalone finite-difference PIDE. Writing $X_t = \log S_t$, the characteristic function factorizes into Heston diffusion and Hawkes jump blocks,

$$\phi_{X_T}(u) = \phi_{\mathrm{diff}}(u)\exp(A(u,T) + B(u,T)\lambda_0), \tag{8}$$

with the jump block determined by the Riccati system

$$\frac{\mathrm{d}B}{\mathrm{d}\tau} = \psi_J(u)e^{\alpha B} - \beta B - iuk - 1, \tag{9}$$

$$\frac{\mathrm{d}A}{\mathrm{d}\tau} = \beta\bar{\lambda}B, \qquad A(0) = B(0) = 0, \tag{10}$$

where $\psi_J(u) = \exp(iu\mu_J - \frac{1}{2}\sigma_J^2 u^2)$. European option prices are recovered by Fourier-COS inversion. The identities $\phi(0) = 1$, the martingale condition at $u = -i$, and the limit $\alpha \to 0$ back to Bates are used as implementation checks.

Model-role constraint

These four processes govern only the gold-price layer. They never generate Barrick production or costs in the integrated valuation.

3.3 Option-surface calibration and predictive test

Every model is fitted to the same structured GLD surface by minimizing a vega-scaled price loss,

$$F(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{C_i^{model}(\theta) - C_i^{mkt}}{\max(\mathrm{Vega}_i, \varepsilon)}\right]^2, \tag{11}$$

which approximates an implied-volatility error without solving an IV inversion inside every optimizer step. Richer models use global Differential Evolution for initialization and constrained SLSQP for local refinement. Feller admissibility is imposed on the square-root variance process, while Hawkes stability requires $n < 1$.

Calibration fit and forecasting are evaluated separately. A date is admitted to the rolling test only when at least 64 eligible observations and three expiries survive the common filters. Each origin uses the fixed CC64 geometry and is scored on the next available dense date. Performance is summarized by

$$R^2_{OOS,m|b} = 1 - \frac{\sum e^2_{m,t,j}}{\sum e^2_{b,t,j}}, \tag{12}$$

against both the origin cross-sectional mean IV and an origin observed-IV surface interpolated in maturity and log-moneyness inside its convex hull. The target spot and rate curve enter only as scoring-date conditioning variables. This design prevents the lowest in-sample IV error from being interpreted automatically as predictive dominance.

3.4 Team 4 cost and production models

For costs, Team 4 first builds a log-space global master chain from mine-level series and then selects an **ARIMA(3,1,0) with drift** by automated information-criterion search. The admitted specification is

$$(1 + 0.9501B + 0.6128B^2 + 0.3003B^3)(1 - B)y_t = 0.0222 + \varepsilon_t. \tag{13}$$

The selected model minimizes both $AIC = -94.64$ and $BIC = -86.87$ (Akaike, 1974; Schwarz, 1978), where

$$AIC = 2k - 2\log\widehat{L}, \qquad BIC = k\log n - 2\log\widehat{L}. \tag{14}$$

Residual diagnostics do not reject white noise or normality in the reported tests, and a 12-period hold-out yields a MAPE of 0.603% with RMSE approximately one third of the naive random-walk benchmark.

The production specifications are not reported as AIC/BIC-selected in the underlying Team 4 study. Instead, production is decomposed physically as

$$Q_t = \mathrm{Ore}_t\,\mathrm{Grade}_t\,\mathrm{Recovery}_t \times 0.91/31.11, \tag{15}$$

with ore processed modelled as $SARIMA(1,0,1)(1,0,0)_4$, average grade as $SARIMA(1,0,0)(0,0,0)_4$, and recovery sampled from bounded historical ranges. The source study also develops Goodman variance propagation, but those operating distributions are not carried into the unified four-engine valuation; only the frozen point vectors are.

3.5 Valuation mapping

Team 5 makes growth internally consistent with reinvestment through

$$b_t = \frac{g_t}{ROIC_t}, \qquad FCFF_t = M_t^{AT} - \max(M_t^{AT}, 0)b_t, \tag{16}$$

where M_t^{AT} is the after-tax component margin. Over the five explicit years, g_t declines linearly from 8.50% to 2.50%, $ROIC_t$ from 14.50% to 10.50%, and the implied reinvestment rate from 58.62% to 23.81%.

The common discount-rate process is mean reverting,

$$w_{t+1} = \bar{w} + (w_t - \bar{w})e^{-\kappa_w} + \sigma_w^* \varepsilon_{t+1}, \tag{17}$$

with $w_0 = 9.10\%$, $\bar{w} = 8.30\%$, $\kappa_w = 0.45$ and annual volatility 0.90%. It is clipped to $[g_\infty + 1.00\%, 25.00\%] = [3.50\%, 25.00\%]$; every gold engine receives the same 8,192-path shock array.

For simulated path i, the common operating contract and discount rates determine enterprise value,

$$EV^{(i)} = \sum_{t=1}^{T} \frac{FCFF_t^{(i)}}{\prod_{s=1}^{t}(1 + WACC_s^{(i)})} + \frac{TV_T^{(i)}}{\prod_{s=1}^{T}(1 + WACC_s^{(i)})}. \tag{18}$$

The terminal block uses $g_\infty = 2.50\%$, $ROIC_\infty = 10.50\%$ and therefore $b_\infty = 23.81\%$:

$$TV_T^{(i)} = \frac{\max(M_T^{AT,(i)}, 0)(1 + g_\infty)(1 - g_\infty/ROIC_\infty)}{WACC_T^{(i)} - g_\infty}, \tag{19}$$

admitted only when $WACC_T^{(i)} > g_\infty$. The present component margin, FCFF and equity bridge remain unreconciled proxies; the final distributions are therefore conditional research sensitivities rather than filing-reconciled fair values.

4 CURRENT OPERATING-MODEL EVIDENCE

The cost and production layers inherited from Team 4 are purpose-built operating forecasts rather than generic statistical add-ons. They were first assembled in an earlier Barrick working paper that used a provisional Black–Scholes/GBM gold-price block. In the unified valuation only the operating method is retained and refreshed with Barrick's Q1–Q2 2026 mine statistics; the legacy gold-price simulation and illustrative EBITDA/valuation are excluded. This keeps the non-price inputs observable while ensuring that all valuation gold paths come from the current Team 8 engines.

4.1 Cost-of-sales block

The cost model in Section 3.4 extracts a common log-space trend, forecasts it with ARIMA(3,1,0) with drift, and reconstructs mine-level paths with volatility-scaled anchors. Its information-criterion selection and hold-out diagnostics support the forecast specification; they do not make CoS a jointly simulated valuation state.

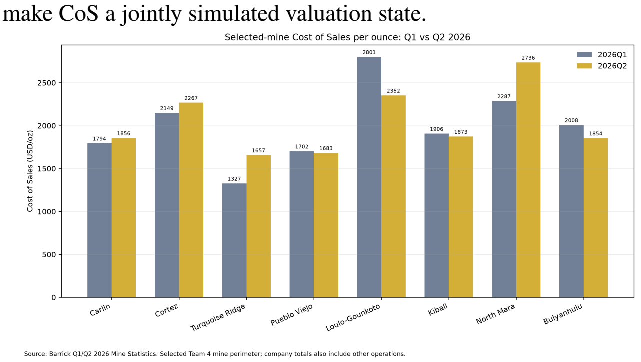


Figure 1. Selected-mine Cost of Sales per ounce in Q1 and Q2 2026. Mine dispersion motivates preserving the Team 4 operating decomposition rather than replacing it with a single company-average cost.

The refreshed actuals provide a transparent starting boundary for the operating forecast. From Q3 2026 onward, the unified experiment accepts the resulting deterministic 20-quarter cost vector rather than the fully stochastic CoS distributions explored in the source study. This design is intentional: simultaneous changes in gold, cost and production uncertainty would prevent clean attribution across the four price engines.

4.2 Production block

Production is reconstructed from ore processed, grade and recovery using the specifications in Section 3.4. This preserves mine heterogeneity. The SARIMA orders are admitted from the source study, which does not report their selection by AIC/BIC.

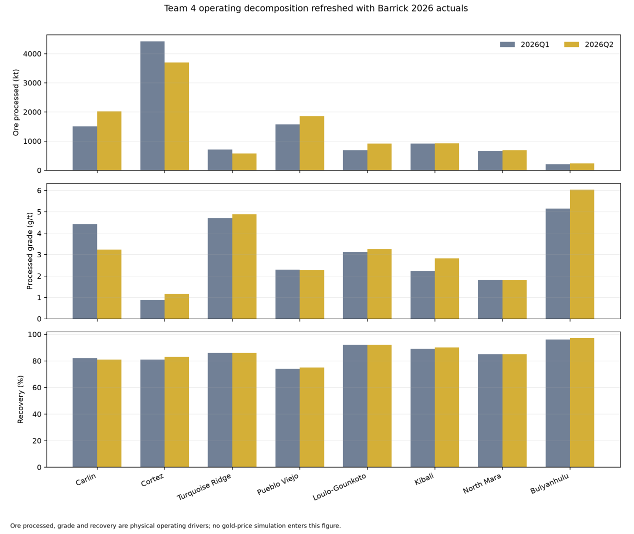


Figure 2. Ore processed, grade and recovery updated with Barrick 2026 actuals. These are physical operating drivers; no commodity-price simulation enters the panel.

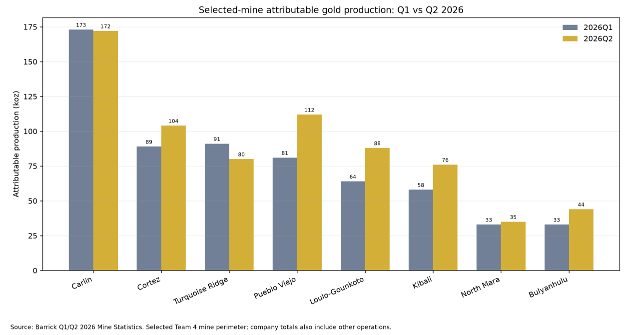


Figure 3. Selected-mine attributable gold production in Q1 and Q2 2026. The current actuals reveal heterogeneous changes that are hidden by a company-level total.

The original study also develops Goodman variance propagation and confidence bands for the production components. Those distributions remain useful evidence on operating-model uncertainty, but they are not propagated jointly in the four-engine corporate experiment: only the refreshed point vectors are passed downstream.

4.3 Forecast handoff used by the valuation

The final handoff keeps observed and modelled quantities distinct. Barrick's Q1–Q2 2026 actual production and CoS establish the current operating state; the Team 4 forecasts begin at Q3 2026 and extend through the common valuation horizon.

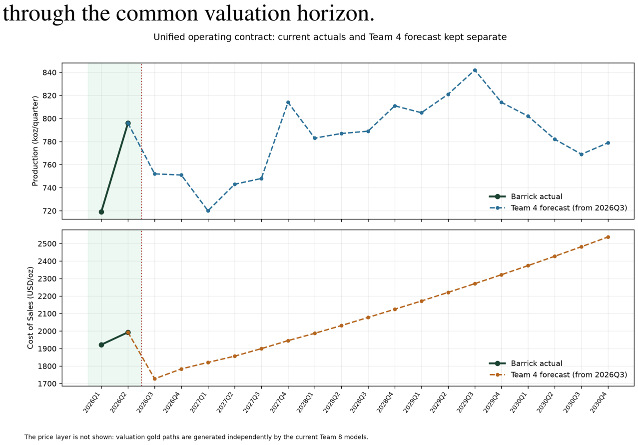


Figure 4. Unified operating contract. Barrick Q1–Q2 2026 actuals are shown separately from the deterministic Team 4 production and CoS vectors used from Q3 2026 onward. The straight cost path is the conditional mean of a persistent-trend specification, not a claim that operating uncertainty disappears.

Prudential operating boundary

Option-implied gold distributions provide forward-looking market information to the revenue layer. Costs retain a persistent trend without additional technology-driven cost reductions. This is prudential relative to otherwise identical efficiency-improvement scenarios, but it does not establish a mathematical lower bound. Stochastic costs could yield favourable and adverse outcomes; the current fixed vectors preserve attribution across gold engines. Growth is already included in the common DCF schedule.

5 CALIBRATION SAMPLE AUDIT

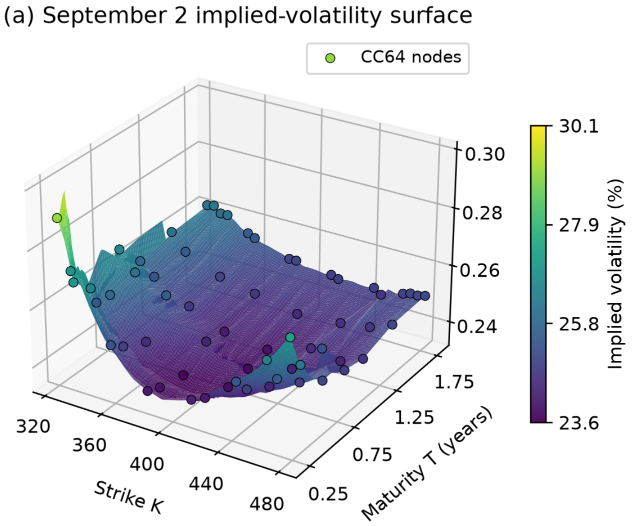


Figure 5. (a) GLD implied-volatility surface, 2 September 2026, with CC64 nodes. Interpolation is used for display only.

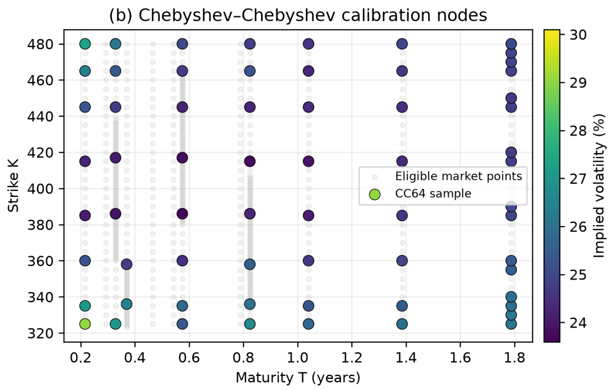


Figure 5 (continued). (b) The 64 actual contracts selected from 605 eligible calls across 12 expiries and 146 strikes.

5.1 Why structured sampling is retained

The option surface is irregular in strike and maturity. Team 8 therefore passes a fixed 8×8 Chebyshev–Chebyshev subset to the optimizer rather than allowing densely listed regions to dominate the loss. Every target is mapped to the nearest still-unused actual option: the 64-point calibration set contains no synthetic contracts. The figure audits the same 2 September parameter snapshot used by the refreshed valuation.

Evidence status

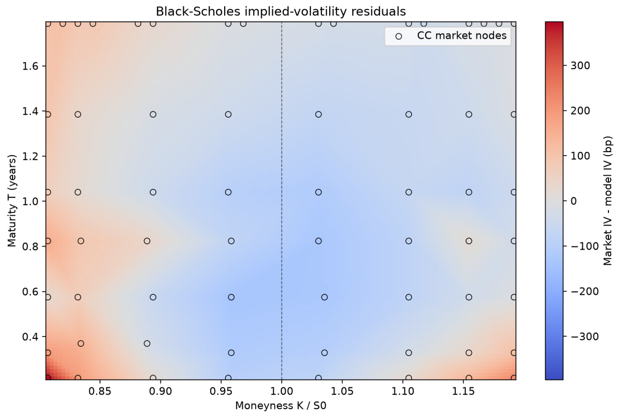
The calibrated inputs, residual maps and valuation paths share the same 2 September Team 8 parameter set. The U.S. Treasury curve is dated 2 September as well; the separate Q2 realized-gold level anchor remains explicit in the valuation contract.

5.2 What the residual maps add

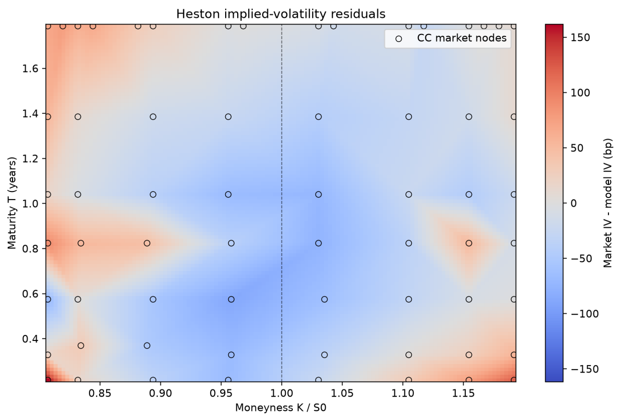
The residual heatmaps show where each specification systematically over- or under-prices the observed implied-volatility surface. Black–Scholes leaves the strongest smile and term-structure pattern. Table 2 compares current-surface fit; Heston has the smallest error. The fitted Hawkes branching ratio is 0.5569, below the stationarity boundary. Common axes support comparison; each colourbar reports its own residual range.

Table 2. Current-surface IV calibration RMSE.

Model	RMSE (bp)
Black–Scholes	97.2107
Heston	49.4881
Bates–Poisson	54.6273
Full Bates–Hawkes	52.9714

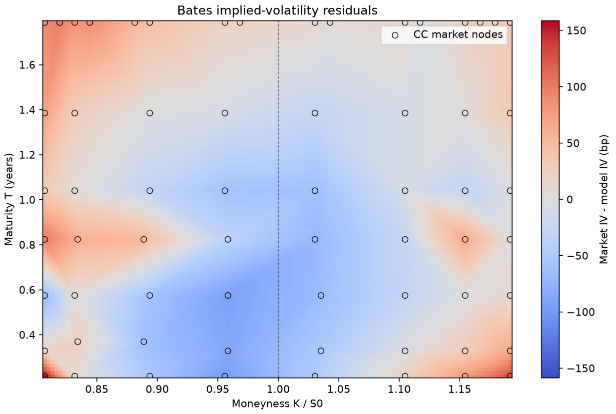


(a) Black–Scholes

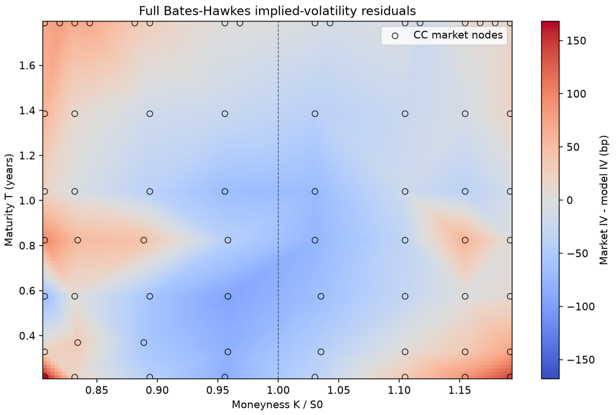


(b) Heston

Figure 6. Implied-volatility residual maps for the common 2 September CC64 sample. Blue denotes model IV above market IV and red denotes model IV below market; Heston records the lowest in-sample IV RMSE.



(c) Bates–Poisson



(d) Full Bates–Hawkes

Figure 6 (continued). Bates–Poisson and Full Bates–Hawkes residual maps; colourbars report model-specific ranges.

6 DCF DRIVER AND TERMINAL-VALUE AUDIT

Growth, reinvestment, WACC and terminal closure are shared across gold engines. Table 3 summarizes the corporate assumptions.

6.1 Growth, ROIC and reinvestment

Growth and ROIC converge linearly to their stable levels over five years. Reinvestment follows $b_t = g_t / ROIC_t$, linking growth to the capital required to sustain it (Figure 7).

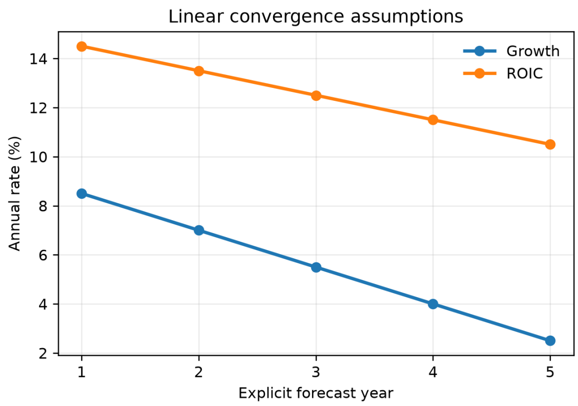


Figure 7. Common explicit-period DCF schedule: growth and ROIC converge linearly over the five-year horizon.

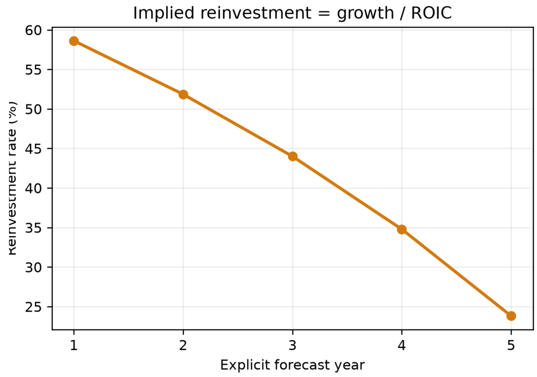


Figure 7 (continued). The implied reinvestment rate, defined as growth divided by ROIC, falls from 58.62% to 23.81%.

6.2 Common stochastic WACC

WACC follows the bounded mean-reverting process in Equation (17), with inputs in Table 3. Its annual conditional standard deviation is 0.7308 percentage points. All branches reuse the same 8,192-path shock array; Figure 8 shows its distribution and zero-shock path.

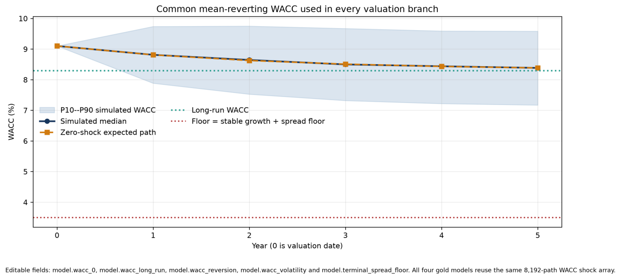


Figure 8. Common mean-reverting WACC. The band is the P10–P90 range from the shared 8,192-path shock array; the orange line is the zero-shock path and the dotted lines show the long-run level and admissibility floor.

6.3 Terminal closure and sensitivity

The terminal block uses $g_\infty = 2.50\%$, $ROIC_\infty = 10.50\%$ and terminal reinvestment of 23.81%. At the current median terminal WACC of 8.384%, the normalized terminal multiple is 13.27× final after-tax margin. The denominator is admitted only when $WACC_T > g_\infty$; the spread floor prevents a mechanically explosive closure in the simulated paths.

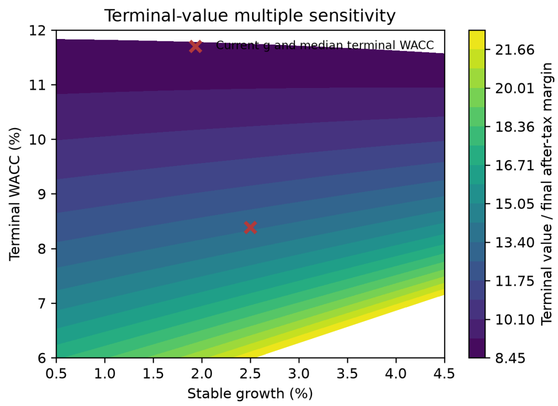


Figure 9. Terminal-value multiple sensitivity to stable growth and terminal WACC. The cross marks the configured stable-growth rate and current median terminal WACC.

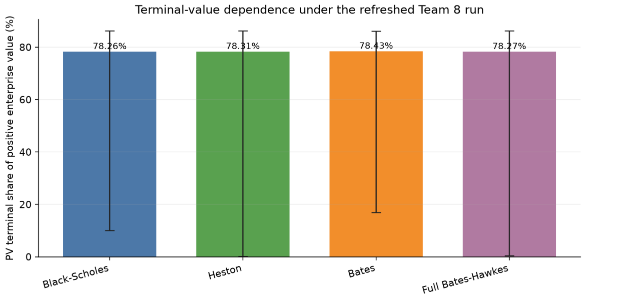


Figure 9 (continued). Present-value terminal share on positive-EV paths. Refreshed medians are 78.26% for Black–Scholes, 78.31% for Heston, 78.43% for Bates and 78.27% for Full Bates–Hawkes; bars show the P10–P90 range.

Table 3. Editable common DCF inputs.

Parameter	Value
Tax rate	25.00%
$g_1 \rightarrow g_\infty$	8.50–2.50%
$ROIC_1 \rightarrow ROIC_\infty$	14.50–10.50%
$WACC_0 \rightarrow \bar{w}$	9.10–8.30%
WACC reversion	0.45
WACC volatility	0.90%
WACC floor/cap	3.50/25.00%
Terminal reinvestment	23.81%

Interpretation

These are transparent methodological assumptions, not filing-calibrated Barrick estimates. Their materiality is visible: the terminal block supplies roughly three quarters of positive enterprise value. They should therefore be changed centrally and sensitivity-tested rather than hidden inside a single headline valuation.

7 CONDITIONAL VALUATION FINDINGS

Table 4. Primary Full Bates–Hawkes signal.

Metric	USD/share or %
P10	3.13
Median (P50)	35.14
P90	104.72
Completed market close	44.13
Market vs. median	+25.58%
Market percentile	59.24th
Paths above market	40.76%

Interpretation

The completed 2 September market close is above the Full Bates–Hawkes conditional median but remains inside a wide right-skewed distribution. The result leans toward overvaluation relative to this model median; it does not establish market mispricing.

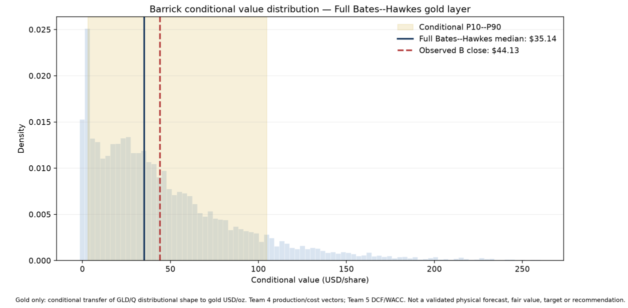


Figure 10. Conditional Barrick value-per-share distribution under the Full Bates–Hawkes gold layer. The reference line is the 2 September 2026 close. The common operating and corporate layers are reused across 8,192 paths; the interval is neither a target-price range nor a confidence interval.

7.1 Cross-model central tendency

The refreshed 2 September run places all four conditional medians in a narrow USD 35.14–35.87 range. Their upper tails nevertheless remain broad: P90 values range from USD 101.43 to USD 104.72. With operations, WACC shocks and the DCF mapping fixed, these differences isolate the distributional effect of the option-implied gold engine.

Table 5. Central comparison across gold engines.

Gold engine	Median	$\Pr(V > 44.13)$
Black–Scholes	35.87	41.92%
Heston	35.67	41.25%
Bates–Poisson	35.84	41.08%
Full Bates–Hawkes	35.14	40.76%

7.2 Why Full Bates–Hawkes is primary

Full Bates–Hawkes is displayed as the primary structural sensitivity because it combines stochastic variance with self-exciting jumps and records the lowest date-equal rolling OOS IV RMSE. This choice is not an in-sample claim: Heston has the lowest current-snapshot IV RMSE. The fitted Hawkes branching ratio of 0.5569 satisfies $n < 1$ and indicates a materially persistent, but stationary, self-excitation channel.

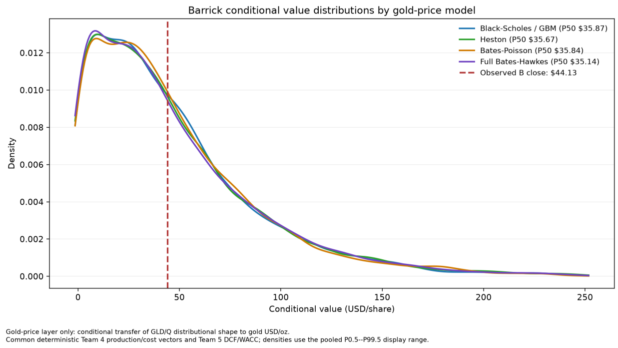


Figure 11. Conditional Barrick value distributions under the four refreshed gold-price engines. The USD 44.13 close is above every median. Operations, WACC and the DCF mapping are common across models.

8 OUT-OF-SAMPLE EVIDENCE AND MODEL RISK

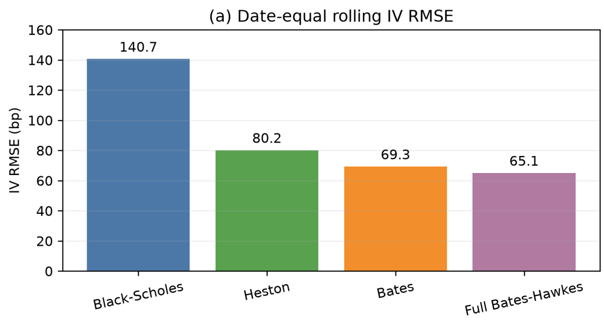


Figure 12. Dense-only rolling evidence on 30 forecast origins and 4,667 common target observations. Full Bates–Hawkes has the lowest date-equal IV RMSE.

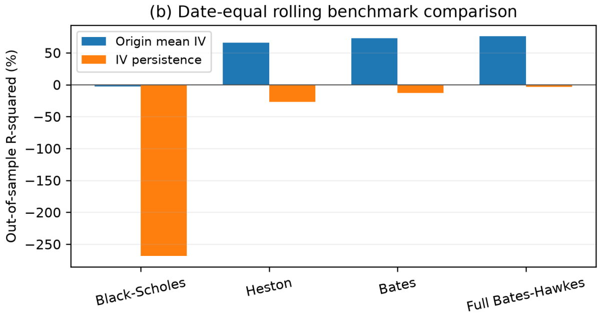


Figure 12 (continued). Positive R^2_{OOS} indicates improvement over the stated benchmark.

8.1 Predictive interpretation

Model comparisons use 4,667 targets; persistence comparisons use the 4,026 inside the origin interpolation domain. Benchmark scores therefore have different supports; equal date weighting limits the influence of densely populated dates.

The rolling exercise separates current-snapshot fit from forward performance. Date-equal IV RMSE is 140.6528 bp for Black–Scholes, 80.1964 bp for Heston, 69.3055 bp for Bates and 65.0507 bp for Full Bates–Hawkes. Relative to the origin mean-IV benchmark, the corresponding R^2_{OOS} values are −2.58%, 66.07%, 72.97% and 75.81%. None beats interpolated IV-surface persistence, but Full Bates–Hawkes is closest at −3.68%, followed by Bates at −12.92% and Heston at −27.24%.

Model-risk conclusion

Heston is the in-sample winner, whereas Full Bates–Hawkes is the rolling OOS winner. The valuation therefore reports all four engines and uses Full Bates–Hawkes only as the primary structural sensitivity, not as a universally dominant forecast.

8.2 Interpretive limitations

The economic reading remains conditional: calibration is under $\mathbb{Q}$ without a validated $\mathbb{Q} \to \mathbb{P}$ map; GLD distributional shape is transferred without a one-for-one conversion to physical gold; the realized-price anchor predates the option snapshot; and the FCFF/equity bridge is not fully filing-reconciled. Debt, cash, minorities, non-operating assets and related bridge adjustments remain explicit zero placeholders. The distributions are research sensitivities, not fair values, confidence intervals or price targets.

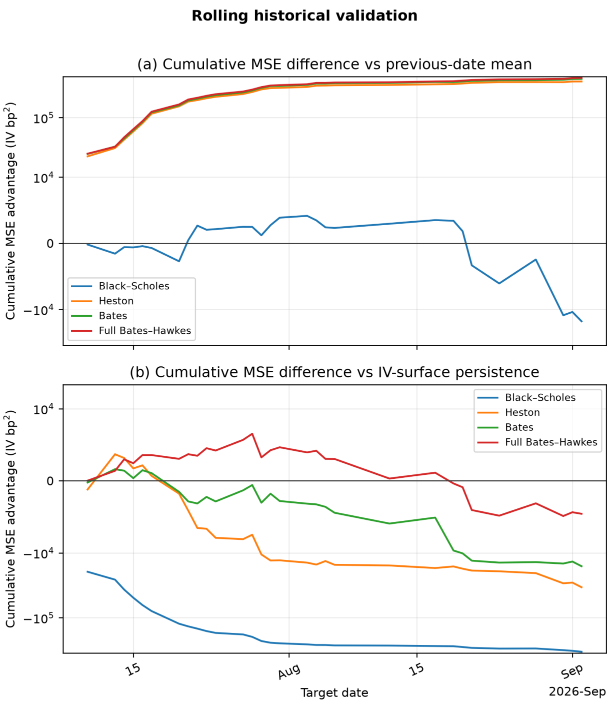


Figure 13. Cumulative rolling MSE differences against the origin mean IV and interpolated IV-surface persistence. Upward movement favours the option model; the stronger persistence benchmark remains difficult to beat.

8.3 Jump-layer interpretation

The self-exciting extension is motivated by clustered option-implied jump activity. Relative to the constant Poisson baseline, the Hawkes specification permits intensity bursts followed by endogenous decay. The current branching ratio is 0.5569: shocks propagate materially, while the stationarity restriction remains satisfied.

Without a dated news filtration or parameter-stability evidence, these option-implied bursts cannot be identified as causal responses to geopolitical events.

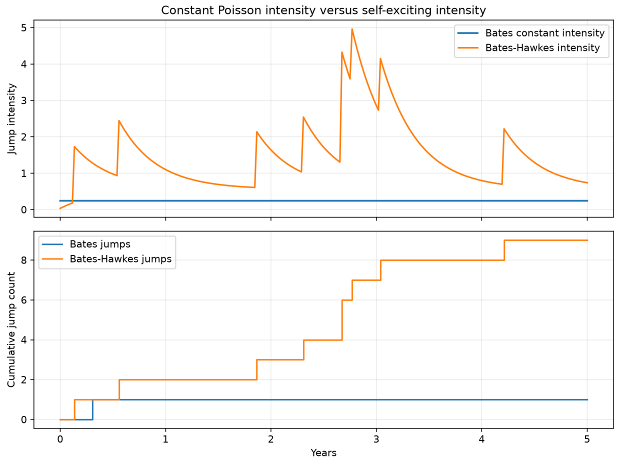


Figure 14. Poisson and Hawkes intensity paths under the 2 September parameter set. The Hawkes layer captures clustered arrivals and decay with a stationary branching ratio of 0.5569.

CONCLUSION

Holding operations and corporate assumptions fixed isolates the effect of gold-price model choice. Heston fits the current surface best; Full Bates–Hawkes leads the rolling OOS comparison, although neither it nor the other engines beats interpolated persistence. The market close exceeds all four conditional medians without establishing market mispricing.

The cost specification is prudential relative to otherwise identical scenarios granting additional technology or efficiency gains. A stochastic extension would expose favourable and adverse cost paths, while its conditional mean could remain smooth. Neither forward-looking option information nor deterministic costs establishes a lower bound on company value. The terminal component, roughly 78% of positive enterprise value, also makes the result sensitive to corporate assumptions.

Decision-relevant takeaway

For a miner, perpetual continuation also requires replacement investment; finite reserve life and closure costs remain unreconciled. The framework supports controlled model comparison. A fair-value interpretation still requires reconciliation of the commodity basis, probability measure, operating uncertainty and equity bridge. Potential productivity gains and their investment costs should be evaluated jointly in future scenarios.

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